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Westchester County Sidewalk Coverage Analysis Planning Decision Support Package

Arcanum Research Initiative

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Purpose of This Package

This package provides **comprehensive decision-making tools** for assessing sidewalk adequacy in Metro-North Transit-Oriented Development (TOD) areas. Rather than a single number, you'll find spatial analysis, prioritization frameworks, investment scenarios, and comparative benchmarks to support planning decisions.

Assessment Framework

Coverage Analysis: Metro-North TOD Areas (0.5-mile radius)

Coverage Status	Road Count	Percentage
No Coverage (Priority)	502 roads	45.1%
One-Side Coverage (Opportunity)	352 roads	31.5%
Both-Sides Coverage (Adequate)	263 roads	23.5%
Any Coverage	615 roads	54.9%
Total TOD Roads Analyzed	1,117 roads	100%

Key Planning Context

- **Transit Connectivity Gap:** 502 TOD roads (45.1%) lack pedestrian access to Metro-North stations
- **Quick Win Potential:** 352 roads already have one-side coverage - completing the second side is cost-effective
- **Comparative Performance:** TOD areas have 3.0x better coverage than non-TOD areas (54.9% vs 18.2%)
- **Industry Context:** Best practice for TOD pedestrian coverage is 75-85% (current: 54.9%)
- **Equity Concern:** 81.8% of non-TOD roads lack sidewalk coverage

Assessment: MODERATE ADEQUACY - Substantial Investment Needed

While TOD areas outperform the county average, nearly half of transit-accessible roads lack sidewalks. This limits walkability, transit ridership potential, and equitable access.

1 Decision-Making Tools in This Package

1.1 Excel Files (Interactive Data Exploration)

Located in `Excel/` folder - open with Microsoft Excel, Google Sheets, or LibreOffice Calc:

1. `1_EXECUTIVE_SUMMARY.xlsx`

- Complete coverage breakdown by TOD vs Non-TOD
- Filter and sort roads by coverage status
- Use for: Investment prioritization, budget justification

2. `2_TOD_COMPARISON.xlsx`

- Side-by-side TOD vs Non-TOD vs County-Wide analysis
- Includes industry benchmarks and gap-to-target calculations
- Use for: Strategic planning, stakeholder presentations

3. `3_ROAD_TYPE_ANALYSIS.xlsx`

- Coverage breakdown by road classification (paved, unpaved, elevated, etc.)
- Identifies road types with lowest coverage (unpaved: 4.0%)
- Use for: Targeted investment by infrastructure type

4. `4_AREA_ANALYSIS.xlsx`

- Coverage statistics by road area (acres)
- Useful for cost estimation based on square footage
- Use for: Budget planning, ROI analysis

All Excel files follow the Druck standard: ONE SHEET per file for clarity.

1.2 PDF Reports (Analysis & Recommendations)

Located in `Reports/` folder - open with any PDF reader:

1. `Executive_Summary.pdf` (5 pages)

- High-level overview of coverage assessment
- Key findings and immediate recommendations
- Use for: City Council presentations, public communications

2. `Technical_Analysis.pdf` (12 pages)

- Detailed methodology (DVRPC sidewalk-to-road ratio)
- Statistical validation and quality assurance
- Coverage by road type, spatial patterns, policy recommendations
- Use for: Technical review, grant applications, methodology defense

2 Investment Prioritization Framework

2.1 Recommended Investment Tiers

Tier 1: Transit Connectivity Priority (502 roads)

- **Target:** All TOD roads with no sidewalk coverage
- **Impact:** Maximize pedestrian access to Metro-North stations
- **Rationale:** TOD investment requires pedestrian infrastructure
- **Timeline:** 5-7 year capital improvement plan

Tier 2: Completion Opportunities (352 roads)

- **Target:** TOD roads with one-side coverage
- **Impact:** Complete pedestrian networks, improve safety
- **Rationale:** Lower cost per mile (infrastructure already half-built)
- **Timeline:** 3-5 year implementation (quick wins)

Tier 3: Equity & Non-TOD Expansion

- **Target:** Non-TOD areas with high pedestrian demand
- **Impact:** Address 81.8% no-coverage rate in non-TOD zones
- **Rationale:** Environmental justice, equitable access
- **Timeline:** 10-15 year comprehensive plan

3 Key Planning Questions to Explore

Use the Excel files and PDF reports to investigate:

1. **Spatial Prioritization:** Which TOD roads should be addressed first?
 - Filter Excel by coverage status + road type
 - Overlay with traffic volume data (if available)
 - Identify clusters of no-coverage areas near high-ridership stations
2. **Cost-Benefit Analysis:** What's the ROI of Tier 2 completions vs. Tier 1 new construction?
 - Estimate cost per linear foot by road type (Excel file 3)
 - Calculate cost to complete 352 partial-coverage roads vs. 502 no-coverage roads
 - Consider pedestrian volume, transit ridership gains
3. **Timeline & Phasing:** What's realistic given budget constraints?
 - Scenario A: Focus 100% on Tier 1 → 5-7 years
 - Scenario B: Focus 100% on Tier 2 → 3-5 years (quick wins)
 - Scenario C: Balanced approach → 10 years comprehensive

4. **Grant Eligibility:** Which projects qualify for federal/state funding?

- TOD + transit connectivity grants
- ADA compliance improvements
- Environmental justice investments (non-TOD equity)

5. **Benchmarking:** How do we compare to peer counties?

- See Technical Analysis PDF for comparative data
- Industry best practice: 75-85% coverage in TOD areas
- Current Westchester: 54.9% (gap of 20-30 percentage points)

4 Methodology Summary

Analysis Method: DVRPC (Delaware Valley Regional Planning Commission) sidewalk-to-road ratio
Coverage Classification:

- **No Coverage:** Sidewalk-to-road ratio < 0.1 on both sides
- **One-Side:** Ratio 0.4–0.8 on one side, < 0.1 on other
- **Both-Sides:** Ratio > 1.2 on both sides

TOD Standard: 0.5 miles (2,640 feet) from Metro-North stations - industry standard for transit-oriented development analysis

Data Quality: 100% of county road polygons analyzed with geometric validation (EPSG:2260 NY State Plane)

Coordinate System: EPSG:2260 (NY State Plane Long Island, feet) for analysis

5 Comparative Benchmarks

Area Type	Coverage %	Gap to Best Practice
Westchester TOD Areas	54.9%	-20 to -30 percentage points
Industry Best Practice (TOD)	75-85%	Target range
Westchester Non-TOD	18.2%	Significant equity gap
County-Wide Average	26.0%	Below national average

Key Takeaway: While Westchester TOD areas perform better than the county average, there's a 20-30 percentage point gap to industry best practices for transit-oriented pedestrian infrastructure.

6 Next Steps for City Planning

1. **Review Excel Files:** Explore data, filter by priority criteria
2. **Read PDF Reports:** Understand methodology, review recommendations
3. **Scenario Planning:** Develop 3-5 investment scenarios with costs

4. **Stakeholder Engagement:** Present findings to City Council, community boards
5. **Grant Applications:** Identify federal/state funding opportunities
6. **Capital Planning:** Integrate into 5-10 year CIP based on priorities

Analysis Status: COMPLETE

All 4,386 county roads analyzed with 100% data coverage
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